

The Importance Of RADIATION PATTERN OF AN ANTENNA

Look beyond an antenna's size and gain; its radiation pattern holds the key to its performance. Understanding this can unlock a whole new perspective on wireless communication



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You need to know more than simply antenna gain to accurately evaluate the performance of antennas. You should know everything about their radiation patterns. Essentially, the gain of an antenna measures the intensity of the beam it produces. Similar to a flashlight, the more focused the beam, the greater its range and the need for accuracy in orientation.

Radiation pattern vs gain

Antenna gain has a straightforward numerical value and is expressed as dBi. The pattern in which an antenna emits radio frequency energy is known as its radiation pattern. The radiation from an antenna depicts how the antenna performs for a specific application in terms of radiation field strength, directivity, gain, phase, and

TABLE 1
RELATION BETWEEN 2D AND 3D

2D Radiation Pattern	3D Radiation Pattern
Resolution is high only in the selected plane	Lower resolution in the entire radiation plane
Projects sharp peaks and nulls in directional antennas	Reveals unexpected pattern and distortion
Radiation efficiency cannot be calculated since entire sphere is not covered	Radiation efficiency vs frequency is possible
Best suited for monopole, log periodic high gain directional antennas	Best suited for PIFA, loop, embedded, and omni directional antennas

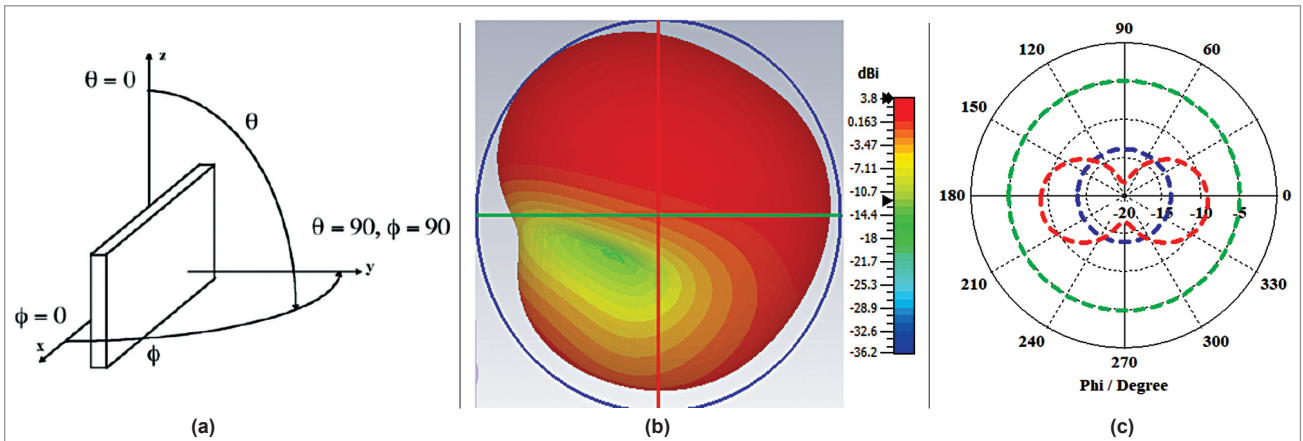


Fig. 1: (a) Coordinate system for an antenna measurement, (b) 3D radiation pattern, and (c) 2D radiation pattern